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FURTHER PALAEOMAGNETIC RESULTS FROM THE FEDERATION OF SOUTH ARABIA

D. H. TARLING, M. SANVER and A. M. J. HUTCHINGS
Department of Geophysics and Planetary Physics, School of Physics, University of Newcastle upon Tyne, England

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Rock samples collected from three separate volcanoes are found to have variable stability of remanent magnetization. Selection of the most stable material confirms previous observations of a westerly declination of remanence in these volcanoes which range in age from 5-10 m.y. This is taken to confirm that the anticlockwise rotation suggested earlier for the Aden volcano is true for South Arabia as a whole, thereby affording further evidence of the dilatational origin of the Red Sea and Gulf of Aden.

1. INTRODUCTION

The recent geological history of the Red Sea area is of crucial importance in geological-geophysical studies as it is regarded by many [1] as marking the initial stages of continental drifting. Irving and Tarling [2] reported palaeomagnetic results from the Aden Volcano in which the directions of remanence were found to deviate by 7 ± 4°W in declination but not in inclination from the axial dipole field. This deviation appeared to be outside experimental error and was consistent with the dilatational opening of the Red Sea subsequent to the formation of the Aden Volcano causing an anticlockwise rotation of Arabia. These early palaeomagnetic results were extremely stable to partial demagnetization by alternating magnetic fields, but were regarded with some suspicion on two points: a) the sample mean remanent directions were extremely close ($\alpha = 2.7^{o}$) corresponding to an abnormally low magnitude of secular variation, and b) at that time the Aden volcano was thought to be of Pleistocene or Quaternary age and this deviation implied that the Red Sea was not in existence about 1 m.y. ago while geological evidence suggests the initiation of faulting between Eocene and Miocene times. It was clearly necessary that further palaeomagnetic work be undertaken, preferably on both sides of the Red Sea. The Royal Society Volcanological Expedition to the Federation of South Arabia [3] provided an opportunity to obtain further samples from Arabia and we are extremely grateful to the leader of the expedition, Dr. I. G. Gass, for allowing one of us (A. M. J. H.) to accompany the expedition and collect the samples on which this report is based.

Samples were obtained from three volcanoes: Jebel Khariz, Little Aden and Aden. As the geology and volcanic history of each of these volcanoes will be described elsewhere [4, 5], only the briefest account of relevant observations is given below. Many of these samples were weathered and were found to be unstable to partial alternating magnetic field demagnetization. Therefore it was necessary to make some selection of the data. This selection was made objectively on the ground of either internal consistency within a single flow or on the laboratory behaviour of specimens to AC demagnetization. The remaining data showed more scatter than had been observed in the previous Aden results but had mean directions which were broadly consistent with the deviation in declination, thereby confirming the rotation of S.W. Arabia.

2. METHOD

The hand samples, collected in the field, were drilled in the laboratory to yield approximately four specimens from each core. The remanence of these specimens was measured using either astatic [6, 7] or spinner [8] magnetometers in which the direction can be determined to $\pm 3^{o}$ and the intensity of remanence to $\pm 5\%$. The susceptibility of each specimen was also determined using a susceptibility bridge [9]. In some sites, all specimens were subjected to partial demagnetization in alternating magnetic fields [10, 11]; in

Table 1

Site mean directions of remanence. N is the number of samples: $\overline{M}$ is the geometric mean intensity of remanence in gauss $\times 10^{-5}$: $\overline{X}$ is the geometric mean susceptibility in gauss oersted$^{-1} \times 10^{-5}$; D, I are the declination and inclination respectively of the mean direction of remanence calculated giving equal weight to each sample direction: O_{63} is the circular standard deviation (radius of cone containing 63% of the observations).

	Site	N	$\overline{M}$	$\overline{X}$	Initial			170 (peak) oersted		
					D	I	O_{63}	D	I	O_{63}
Jebel Khariz 12.7°N, 44.1°E										
Locality 1 (750 m N.E. of Huwayrib)	1	5	22	9	351 *	-1	48	345	+15	40
	2	3	12	12	353	-8	23	341	-5	11
	3	2	21	7	336 **	-16	29	4	-14	6
	4	3	14	36	12	+7	9	18 **	-13	43
	5	4	4	1	354	-22	6	354	+23	5
Locality 2 ($5\frac{1}{2}$ km E. of Huwayrib)	6	3	10	14	342	+15	9	352	+12	16
	7	2	9	33	357	+4	2	353 *	-31	16
	8	3	15	26	358	+16	21	349	+21	18
	9	2	6	8	345 *	+15	14	309 **	+27	33
	10	5	16	54	338	+7	25	304	+29	32
	11	3	37	9	271 *	-18	31	314	+24	7
Locality 3 (22 km N. of Khor Winerah)	12	3	771	3	161 *	+18	60	180 **	+32	51
	13	3	1	6	189	-11	4	1	-8	18
	14	3	1	5	5	-4	15	0	-5	20
	15	4	42	5	346	-1	6	323	+34	25
	16	3	17	3	10	+26	12	33 **	+51	48
	17	3	6	10	2	+15	9	33 **	+4	41
	18	3	7	14	357	+8	7	10 *	+16	31
	19	3	5	9	358	+14	23	359 **	+46	45
	20	3	18	8	0	+3	5	18	+39	38
Locality 4 (Post caldera collapse)	21	3	21	46	357	+12	3	172	+24	12
	22	3	13	44	359	+15	9	358	+14	5
	23	3	8	47	21 **	0	37	17 **	-15	39
Little Aden Volcano 12.7°N, 44.9°E										
Locality 1	36	2	52	30	90 **	+28	47	144 **	-4	20
	37	4	38	41	234	-1	6	233	-22	23
	38	3	11	58	171	-23	5	176	-25	11
Locality 2	39	3	127	5	147 **	-51	57	136 **	-36	57
	40	3	13	7	223	-10	13	174	-4	12
	41†	3	5	10	196	-12	19	187	-15	14
	42†	3	2	11	155 **	-26	14	195	-27	11
Locality 3	43	1	5	12	347	-4	-	52	-21	-
Aden Volcano 12.8°N, 44.9°E	24†	3	1	12	313 **	-1	55	0	+23	5
	25†	2	2	17	110 **	-33	57	7 **	+5	26
	26	3	6	7	321	+9	8	36 **	+21	36
	27	3	1	0.3	0	+16	9	1	+18	7
	28†	3	2	15	9	+2	17	253 **	+65	95
	29†	3	3	7	175	-1	12	115 **	-4	53
	30	3	281	19	347	-15	2	37 **	-23	56
	31	2	6	10	188	-8	7	192 *	-14	14
	32	2	4	5	217 *	+12	17	208 *	-12	11
	33†	4	4	10	348 **	+27	66	72 **	+8	69
	34	3	5	6	355	+31	7	339 **	+17	42
	35	3	3	4	356	+11	4	28 **	+9	39

* Not significant at a 99% level.
** Not significant at a 95% level.
† Intrusive.

other sites one or two representative pilot specimens were selected and subjected to identical partial demagnetization. All remaining specimens were then treated in 170 (peak) oersted alternating magnetic field in order to reduce the effect of low coercivity components. The directions of remanence were meaned [12], combining specimens to obtain sample mean directions, and combining sample directions to obtain site mean directions. The mean intensities and susceptibilities for each site were determined assuming a log-normal distribution [13] and giving equal weight to each specimen observation.

The palaeomagnetic results from localities in each volcano are described separately below, followed by a discussion of palaeomagnetic properties of the samples from this region. The site mean data are tabulated in table 1 and plotted in fig. 1.

3. RESULTS

3.1. *The Jebel Khariz Volcano*

This volcano was sampled at four localities (table 1). Most of the sites showed some degree of weathering although the freshest material was sampled in each case. The two oldest localities (1 and 3) were in lavas on the flanks of the volcano while locality 3 was in lavas and ignimbrites associated with a side vent. The youngest locality (4) was near the base of lavas erupted following caldera collapse of the main vent. One of the youngest lavas of this volcano gives a K/Ar age of about 10 m.y. [4].

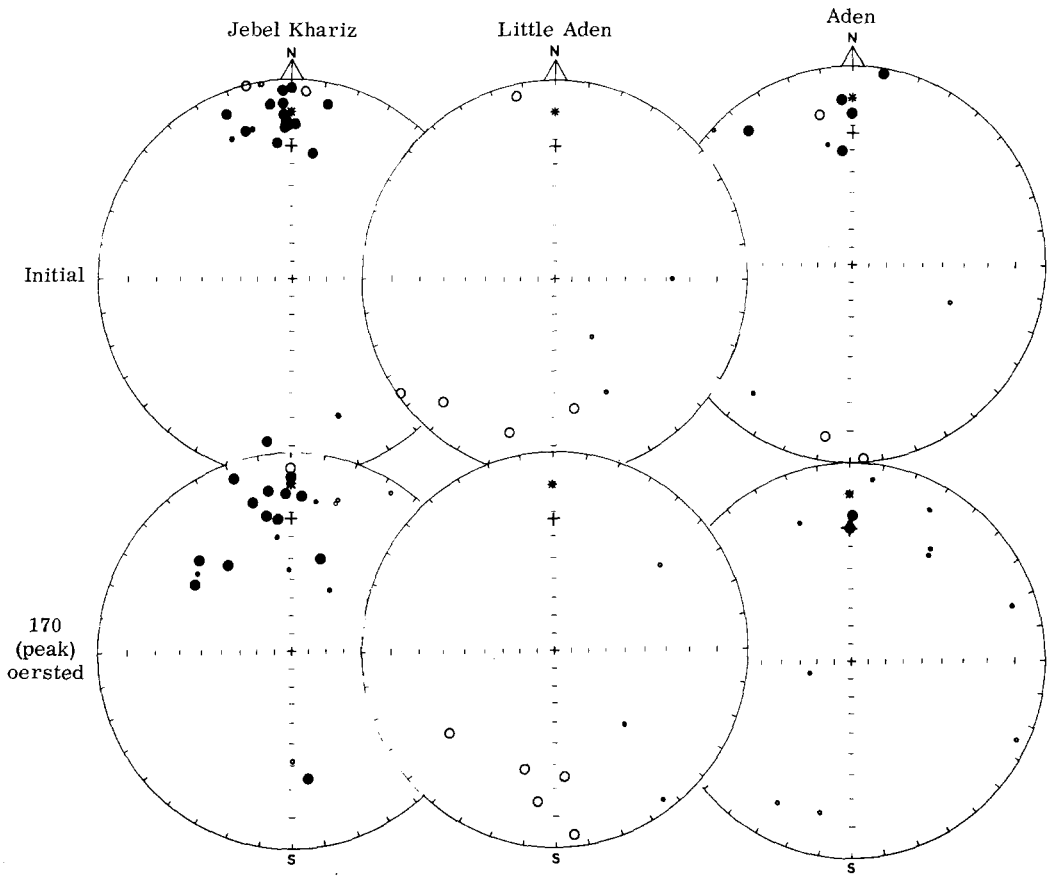


Fig. 1. Mean site direction of remanence before and after partial demagnetization in 170 (peak) oersted. The directions are plotted on an equal angle (Wulff) projection in which the circumference is horizontal. Large symbols are site mean directions which are significant at a 99% level of significance [15, 16]. Hollow circles represent negative downward inclinations and dots represent positive inclinations. The axial dipole field direction is represented by a cross and the present earth's magnetic direction is shown as a star.

In locality 1 all the sample directions were of normal polarity. The initial sample directions in sites 1 and 3 were very scattered, although this scatter was reduced by partial demagnetization in 170 (peak) oersted; however, this treatment considerably increased the sample direction scatter within site 4. Alternating magnetic field demagnetization of pilot specimens showed generally stable remanence, although there was often considerable variation of stability within the same site.

The initial remanent directions at locality 2 were also of normal polarity although the sample directions in sites 9 and 11 were scattered. Treatment in 170 (peak) oersted reduced the sample direction scatter in site 11, but increased the scatter in sites 7 and 9. Alternating magnetic field demagnetization of pilot specimens showed generally poor stability although most sites contained a few pilot specimens with stable remanence.

Locality 3 contained normal and reversed polarities among the initial directions. However, most sample directions became normal after treatment in 170 (peak) oersted although the scatter of sample directions within sites 12 and 16-19 became very large after this treatment. Demagnetization of pilot specimens showed instability in most sites, except site 12; however, the directions of remanence at successive alternating magnetic fields showed considerable scatter between different samples.

The fourth locality had normal initial directions of remanence although the sample directions at site 23 were scattered. Treatment in 170 (peak) oersted did not improve the scatter in site 23, but the remanent polarity became approximately reversed in site 21 (which overlies site 22). The pilot specimen from site 21 showed stable remanence.

3.2. *The Little Aden Volcano*

This volcano was sampled at three localities. The first locality was in the youngest welded ash and lavas lying approximately horizontal. The second locality consisted of lavas and dykes on the southeastern flank and the third locality was on the main ridge of the volcano. All the samples were taken from weathered outcrops although the freshest material was sampled. Radiometric data [5] indicate that this volcano was active between 5 and 6 m.y. ago.

The initial directions in locality 1 were mostly reversed both before and after treatment in 170 (peak) oersted alternating magnetic field. This treatment reduced the scatter in site 36 but the sample directions were still widely scattered. Demagnetization of the pilot specimens showed both sites 36 and 37 to contain unstable specimens.

Locality 2 also contained mainly reversed sample directions both before and after treatment in 170 (peak) oersted. This treatment reduced the scatter in site 42 but the sample directions remained scattered in site 39. Pilot specimen demagnetization showed only poor stability in all sites.

Only one sample was obtained in locality 3 and this sample showed considerable variation in specimen directions both before and after treatment in 170 (peak) oersted, and the pilot specimen showed instability.

3.3. *The Aden Volcano*

Twelve sites were sampled in this volcano, five in dykes and the remainder in lavas. The flows are about 5 m.y. old [5]. Many of the sites were in weathered rock although some samples were obtained from comparatively fresh exposures in quarries and road cuttings. The initial sample directions were somewhat scattered although there was a tendency to cluster into normal and reversed groups. Treatment in 170 (peak) oersted increased the scatter of sample directions within most sites and the grouping of reversed directions of remanence became even less defined. Alternating magnetic field demagnetization of pilot specimens showed generally poor stability although two sites contained specimens with stable remanence.

3.4. *The Aden Younger volcanics*

Many of the samples from all three volcanoes were known to be weathered and it is thought that the generally poor stability (fig. 2), indicated by the behaviour of pilot specimens to partial demagnetization and by the generally large within-site scatter of sample directions, is related to the effects of weathering and, in some cases, lightning. It is clear, therefore, that some selection is necessary to eliminate unreliable directions of remanence in order to isolate, if possible, the original direction of the earth's magnetic field at the time that the volcanoes were active. Two criteria have been used to select data, irrespective of polarity, viz.: a) small scatter of sample directions within a site; the site mean direction had to be significant at the 99% level [15, 16], or b) good stability of remanence directions of pilot specimens assessed by a stability index of 2.5 or greater. These se-

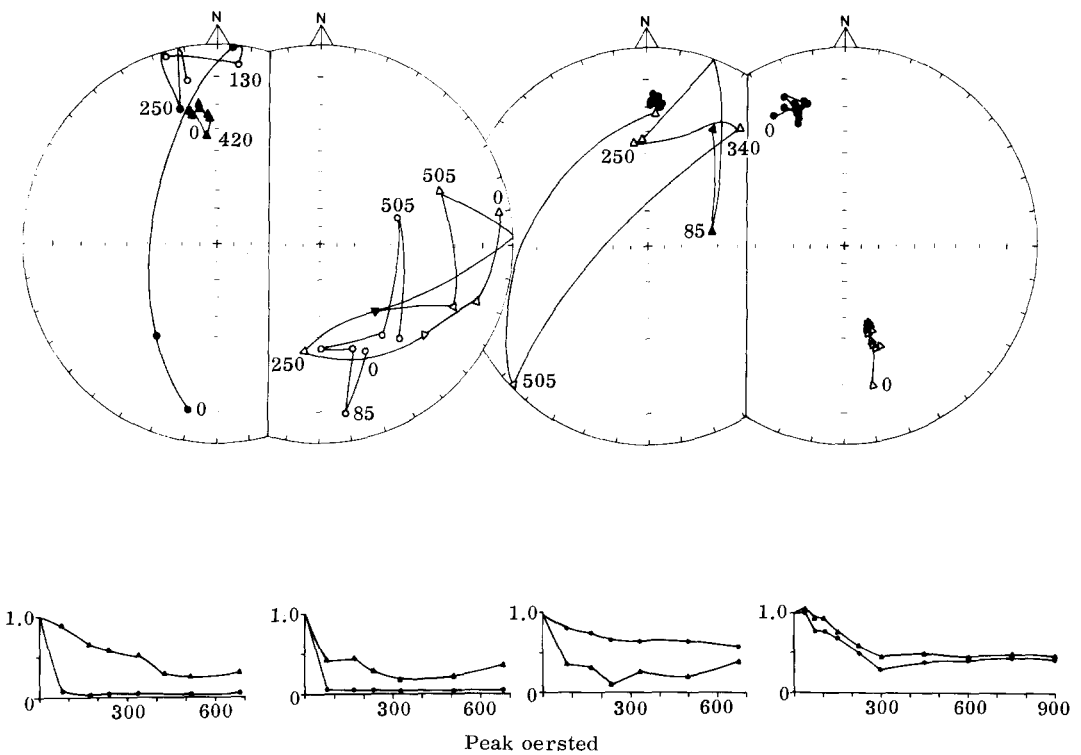


Fig. 2. Examples of alternating magnetic field demagnetization behaviour. The changes in direction at successive increments of field are shown on Wulff projections with the circumference as horizontal: connecting lines do not necessarily represent the actual path of movement of the remanent vector between successive treatments. The ratio of the observed to initial intensity is plotted beneath each stereonet for each pair of examples. The relevant statistical data are tabulated below:

Site	Specimens	Declination	Inclination	α_{95}	Stability index	Most stable range
Jebel Khariz						
Δ 5	34b2	351	+22	1	20.3	50-170
○ 13	45a1	354	+1	16	1.0	85-340
Little Aden						
○ 39	118a1	162	-28	15	1.2	0-675
Δ 36	112a2	114	-12	37	0.6	0-675
Aden (1964 collection)						
○ 27	64a1	4	+17	2	7.7	0-250
Δ 33	105a2	11	-29	40	0.5	170-340
Aden (1961) collection)						
Δ -	31a1	170	-31	1	10.1	40-110
○ -	2a1	340	+18	2	7.2	40-900

Declination, inclination, are the mean declination and inclination of the most stable remanent vector [14] and α_{95} is the radius of a core of 95% confidence of the directions in the most stable range (peak oersted).

Table 2

The data are calculated on site mean values where N is the number of sites. D and I are the mean declination and inclination, calculated irrespective of sign. O_{63} is the circular standard deviation and SI is the geometric mean stability index. The data are selected using: A) All site mean values after treatment in 170 (peak) oersted except Aden 1961 where initial site values are used. B) All sites in A with a mean direction significant at a 99% level ($P = 0.01$). C) All most stable directions of pilot specimens subjected to step demagnetization. D) All site data in C calculated on pilot specimens with a stability index greater than 2.5. The rotation, positive is anticlockwise, is that assumed if the original direction was that of the axial dipole field (0, +24).

Volcano	Selection	N	D	I	O_{63}	SI	Rotation
Jebel Khariz *	A	23	354	+21	31	-	6 ± 6
	B	14	347	+22	25	-	13 ± 7
	C	17	351	+19	16	2.3	9 ± 4
	D	12	345	+21	19	4.8	15 ± 6
Little Aden	A	8	5	+17	38	-	-5 ± 13
	B	5	12	+20	24	-	-12 ± 11
	C	6	337	+40	55	1.3	23 ± 22
	D	0	-	-	-	-	-
Aden (1964)	A	12	11	+18	42	-	-11 ± 12
	B	2	0	+21	4	-	0 ± 2
	C	6	6	+8	30	1.6	-6 ± 12
	D	2	5	+18	2	4.6	-5 ± 1
Aden (1961)	A	12	353	+23	9	-	7 ± 2
	B	9	354	+23	7	-	6 ± 2
	C	8	354	+26	13	7.4	6 ± 5
	D	7	353	+26	14	9.2	7 ± 5
Aden (combined)	A	24	358	+22	38	-	2 ± 8
	B	11	355	+22	7	-	5 ± 2
	C	14	359	+19	23	3.9	1 ± 6
	D	9	356	+24	13	8.4	4 ± 4

* Mean directions of Jebel Khariz site are corrected for a maximum geological estimate of tilt (6° to south); this correction only significantly effects the inclination.

lected data are presented in table 2 which also includes the rejected data.

The data from the Jebel Khariz Volcano show quite good agreement in direction between the rejected and selected data, with declination consistently west of north and inclination shallower than the axial dipole field in Aden (±24°). When the mean directions are corrected for the maximum estimated tectonic tilt since the formation of the volcano (6° to the south) [4] the mean declination is little changed and the mean inclination is much closer to that of the axial dipole field.

The Little Aden data show conflicting results, the declination based on site values being east of north, and west of north based on the pilot specimen data. However, in view of the absence of stable pilot specimens from this volcano, it is not considered that the remanent directions are sufficiently reliable for interpretation in terms of small scale rotations.

The results determined for the Aden Volcano are in apparent conflict with those determined previously [2] as the new declination observed is east of north compared with a previously consistent direction west of north. However, the number of reliable observations in the 1964 collection is much lower than those of the 1961 collection; only two site directions are significant at a 99% level and only three pilot specimens are stable to AC demagnetization. The combined Aden data, therefore, continue to yield a westerly declination while the inclination still agrees with the axial dipole field.

4. CONCLUSIONS

The samples collected most recently from the South Arabian Federation generally exhibit poor stability in alternating magnetic fields and if unstable data are excluded, the remaining information is imprecise for exact determination of the degree of rotation of Southern Arabia. However, a westerly declination is observed for rocks from both Jebel Khariz and Aden volcanoes (table 2). This rotation is therefore consistent with that previously observed for samples from the Aden Volcano alone. The more westerly declination of the Jebel Khariz remanence, when com-

pared with that of the Aden Volcano, may imply that the Red Sea and Gulf of Aden were already opening between 10 and 5 m.y. ago; however, the between-site scatter is quite large and the Jebel Khariz mean direction is not significantly different from that of the Aden Volcano and the complete dilation may have taken place during the last 5 m.y. alone. The easterly declination observed in rocks from Little Aden is not thought to be significant as it does not depart significantly from the axial dipole direction, it is calculated on few data and all specimens tested for stability in alternating magnetic fields proved to be unstable or metastable.

The scatter of between-site directions probably reflects differing stability rather than secular variations of the earth's magnetic field; nevertheless the scatter attributable to secular variation is much larger than was originally observed in the Aden Volcano.

These new data, therefore, support the original view that westerly declinations observed in the remanence of rocks in the Aden Newer Volcanics were more consistent with a rotation of the Arabian mainland than with small scale local rotations. This is, of course, consistent with other geological-geophysical observations [17]. However, the palaeomagnetic data from Arabia must still be regarded as unsatisfactory for precise determination of the rotation of Arabia and results are still required from the African side of the Gulf of Aden, although results from Kenya of Miocene and Plio-Pleistocene age [18] suggest that Africa has remained stationary during the time that Arabia has rotated.

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